

# Nonparametric Estimation

## Chapter 14 – Additive model

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## Structured nonparametric models

We have observed that for multivariate NW and locally linear case,  $(\mathbf{X} \in \mathbb{R}^d, Y \in \mathbb{R})$

$$\text{MSE}(\hat{m}(\mathbf{x})|\mathbf{X}_1, \dots, \mathbf{X}_n) = O_p(n^{-4/(d+4)})$$

To alleviate curse of dimensionality, postulate

$$m = G(\theta, f_1, \dots, f_r)$$

for some known function  $G$  and unknown nonparametric  $f_j : [0, 1]^{d_j} \rightarrow \mathbb{R}$  where  $d_j < d$  and unknown parameter  $\theta \in \mathbb{R}^k$ .

# Additive Models

For  $\mathbf{X}_i \in \mathbb{R}^d, Y_i \in \mathbb{R}$ , iid  $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n)$ .

An additive model for  $m : [0, 1]^d \rightarrow \mathbb{R}$ ,  $m(\mathbf{x}) = \mathbb{E}[Y|\mathbf{X} = \mathbf{x}] = m_1(x_1) + \dots + m_d(x_d)$  where each  $m_j$  is a univariate function.

Then we get

$$\mathbb{E}(Y|X_j = x_j) = \mathbb{E}(m(\mathbf{x})|X_j = x_j) = m_j(x_j) + \sum_{k \neq j}^d \mathbb{E}(m_k(x_k)|X_j = x_j) \quad 1 \leq j \leq d$$

since  $\mathbb{E}(\epsilon|X_j) = \mathbb{E}[\mathbb{E}[\epsilon|\mathbf{X}]|X_j] = 0$ .

## Additive models

Let  $p_j$  for  $1 \leq j \leq d$  and  $p_{j,k}$  for  $1 \leq j \neq k \leq d$  denote the marginal density functions of  $X_j$ , two dimensional density functions of  $(X_j, X_k)$ .

Also, let  $f_j : [0, 1] \rightarrow \mathbb{R}$  be the marginal regression function defined by

$$f_j(\cdot) = \mathbb{E}(Y|X_j = \cdot), \quad 1 \leq j \leq d$$

Then we get

$$\begin{aligned} f_j(x_j) &= m_j(x_j) + \sum_{k \neq j}^d \int_{[0,1]^{d-1}} m_k(x_k) \frac{p(x_1, \dots, x_d)}{p_j(x_j)} d\mathbf{x}_{-j} \\ &= m_j(x_j) + \sum_{k \neq j}^d \int_0^1 m_k(x_k) \frac{p_{j,k}(x_j, x_k)}{p_j(x_j)} dx_k, \quad 1 \leq j \leq d \end{aligned}$$

## Normalized kernel

We take a baseline kernel  $K : [-1, 1] \rightarrow [0, \infty)$  that is bounded and  $\int K = 1$ .

**Normalized kernel.** For  $h > 0$ , define  $K_h : [0, 1] \times [0, 1] \rightarrow [0, \infty)$  by

$$K_h(x, u) = \frac{K(\frac{x-u}{h})}{\int_0^1 K(\frac{v-u}{h})dv} = \frac{K(\frac{x-u}{h})}{h \int_{-u/h}^{(1-u)/h} K(t)dt}$$

Then we get following properties

$$\int_0^1 K_h(x, u)dx = 1, \text{ for all } u \in [0, 1].$$

$$K_h(x, u) = \frac{K(\frac{x-u}{h})}{h} = K_h(x - u) \text{ for all } u \in [0, 1] \text{ if } x \in [2h, 1 - 2h].$$

since  $K(\frac{x-u}{h}) = 0$  if  $u \notin [h, 1 - h]$ ,  $\int_0^1 K(\frac{v-u}{h})dv = h$  if  $u \in [h, 1 - h]$ .

## KDEs, marginal regression estimators

Let  $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n)$  be the observed  $(\mathbf{X}, Y)$ . Put  $\mathbf{X}_i = (X_{i1}, \dots, X_{id})^\top$ . Using the normalized kernel functions,  $K_{h_j}(\cdot, \cdot)$  with different sizes of bandwidth for different coordinate directions, we estimate  $p_j, p_{j,k}, p$  and  $f_j$  by

$$\hat{p}_j(x_j) = n^{-1} \sum_{i=1}^n K_{h_j}(x_j, X_{ij}), \quad 1 \leq j \leq d,$$

$$\hat{p}_{j,k}(x_j, x_k) = n^{-1} \sum_{i=1}^n K_{h_j}(x_j, X_{ij}) K_{h_k}(x_k, X_{ik}), \quad 1 \leq j \neq k \leq d,$$

$$\hat{p}(\mathbf{x}) = n^{-1} \sum_{i=1}^n \prod_{\ell=1}^d K_{h_\ell}(x_\ell, X_{i\ell}),$$

$$\hat{f}_j(x_j) = \hat{p}_j(x_j)^{-1} \cdot n^{-1} \sum_{i=1}^n K_{h_j}(x_j, X_{ij}) Y_i, \quad 1 \leq j \leq d.$$

# Marginalization property of KDEs

Because of the normalization property, we get

$$\int_{[0,1]^{d-2}} \hat{p}(\mathbf{x}) d\mathbf{x}_{-(j,k)} = \hat{p}_{j,k}(x_j, x_k),$$

$$\int_0^1 \hat{p}_{j,k}(x_j, x_k) dx_k = \hat{p}_j(x_j),$$

$$\int_0^1 \hat{p}_j(x_j) dx_j = 1,$$

where  $\mathbf{x}_{-(j,k)}$  is the  $(d-2)$ -vector obtained from  $\mathbf{x} = (x_1, \dots, x_d)^\top$  by omitting  $x_j$  and  $x_k$ , and  $\mathbf{x}_{-j}$  is the  $d-1$ -vector from  $\mathbf{x} = (x_1, \dots, x_d)^\top$  by omitting  $x_j$ .

# SBF estimator

**Basic Assumptions.** For all  $x_j \in [0, 1]$ ,  $1 \leq j \leq d$  and  $1 \leq j \neq k \leq d$ ,

$$\hat{p}_j(x_j) > 0, \quad \int_0^1 \frac{\hat{p}_{j,k}(x_j, x_k)^2}{\hat{p}_k(x_k)} dx_k < \infty$$

**The SBF equation.** Plugging  $\hat{f}_j, \hat{p}_j, \hat{p}_{j,k}$

$$\hat{m}_j(x_j) = \hat{f}_j(x_j) + \sum_{k \neq j}^d \int_0^1 \hat{m}_k(x_k) \frac{\hat{p}_{j,k}(x_j, x_k)}{\hat{p}_j(x_j)} dx_k, \quad 1 \leq j \leq d$$

The integrals exist if  $\int_0^1 \hat{m}_j(x_j)^2 \hat{p}_j(x_j) dx_j < \infty$ ,  $1 \leq j \leq d$  since

$$\left( \int_0^1 \hat{m}_k(x_k) \frac{\hat{p}_{j,k}(x_j, x_k)}{\hat{p}_j(x_j)} dx_k \right)^2 \leq \int_0^1 \hat{m}_k(x_k)^2 \hat{p}_k(x_k) dx_k \cdot \int_0^1 \frac{\hat{p}_{j,k}(x_j, x_k)^2}{\hat{p}_j(x_j)^2 \hat{p}_k(x_k)} dx_k$$

for all  $x_j$ .



# Existence of integrals

$$\int_0^1 \hat{m}_j(x_j)^2 \hat{p}_j(x_j) dx_j = \int_{[0,1]^d} \hat{m}_j(x_j)^2 \hat{p}(\mathbf{x}) d\mathbf{x} = \|\hat{m}_j\|_{2,n}^2$$

The SBF equations determines the sum  $\hat{m} : [0, 1]^d \rightarrow \mathbb{R}$  **uniquely** if we restrict  $\hat{m} \in \mathcal{H}_+$ , basic assumptions and the following holds.

$$\int_{[0,1]^2} \frac{\hat{p}_{j,k}(x_j, x_k)^2}{\hat{p}_j(x_j) \hat{p}_k(x_k)} dx_j dx_k < \infty$$

## Restricting function space

- Consider  $L^2([0, 1]^d, \mu_n)$  with probability measure  $\mu_n$  such that  $\int f(\mathbf{x}) d\mu_n(\mathbf{x}) = \int f(\mathbf{x}) \hat{p}(\mathbf{x}) d\mathbf{x}$ .  $L^2([0, 1]^d, \mu_n)$  is a space of functions  $g : [0, 1]^d \rightarrow \mathbb{R}$  such that

$$\int_{[0,1]^d} g(\mathbf{x})^2 \hat{p}(\mathbf{x}) d\mathbf{x} < \infty.$$

and let  $\|g\|_{2,n}^2 = \int_{[0,1]^d} g(\mathbf{x})^2 \hat{p}(\mathbf{x}) d\mathbf{x}$ .

- Let  $\mathcal{H}_j$  for  $1 \leq j \leq d$  be subspaces of  $L^2([0, 1]^d, \mu_n)$ :

$$g \in \mathcal{H}_j \text{ iff } g(\mathbf{x}) = g_j(x_j) \text{ for some univariate function } g_j.$$

- Let  $\mathcal{H}_+ = \{g | g(\mathbf{x}) = g_1(x_1) + \cdots + g_d(x_d) \text{ with } g_j \in \mathcal{H}_j, 1 \leq j \leq d\}$ . Since  $\|g\|_{2,n} \leq \|g_1\|_{2,n} + \cdots + \|g_d\|_{2,n} < \infty$ ,  $\mathcal{H}_+$  is subspace of  $L^2([0, 1]^d, \mu_n)$ .

$$\mathcal{H}_j \subset \mathcal{H}_+ \subset L^2([0, 1]^d, \mu_n)$$

## Uniqueness of $\hat{m}_j$

If  $\hat{m}(\mathbf{x}) = \hat{m}'(\mathbf{x})$ , then

$$(\hat{m}(x_1) - \hat{m}'(x_1)) + \cdots + (\hat{m}(x_d) - \hat{m}'(x_d)) = 0$$

for  $\mathbf{x} \in \mathbb{R}^d$ .

$(\hat{m}_1 + c_1, \hat{m}_2 + c_2, \dots, \hat{m}_{d-1} + c_{d-1}, m_d - (c_1 + \dots + c_{d-1}))$  also satisfies

$$\hat{m}_j(x_j) = \hat{f}_j(x_j) + \sum_{k \neq j}^d \int_0^1 \hat{m}_k(x_k) \frac{\hat{p}_{j,k}(x_j, x_k)}{\hat{p}_j(x_j)} dx_k, \quad 1 \leq j \leq d$$

since  $\int_0^1 \frac{\hat{p}_{j,k}(x_j, x_k)}{\hat{p}_j(x_j)} dx_k = 1$ .

To determine each  $\hat{m}_j$ , impose following constraints

$$\int_0^1 \hat{m}_j(x_j) \hat{p}(x_j) dx_j = 0, \quad 1 \leq j \leq d$$

## Alternative derivation of SBF equation

Define  $\mathcal{F} : \mathcal{H}_+ \rightarrow [0, \infty)$  by

$$\mathcal{F}(g) = \int_{[0,1]^d} \left( n^{-1} \sum_{i=1}^n \left( Y_i - g_1(x_1) - \cdots - g_d(x_d) \right)^2 \right) \prod_{j=1}^d K_h(x_j, X_{ij}) d\mathbf{x}.$$

Then **Fréchet (Gâteaux) derivatives of  $\mathcal{F}$** : For a 'direction'  $\delta \in \mathcal{H}_+$ ,

$$\mathcal{F}'(g)(\delta) = -2 \int_{[0,1]^d} n^{-1} \sum_{i=1}^n (Y_i - g_1(x_1) - \cdots - g_d(x_d)) \delta(\mathbf{x}) \prod_{j=1}^d K_{h_j}(x_j, X_{ij}) d\mathbf{x}.$$

since

$$|\mathcal{F}(g + \delta) - \mathcal{F}(g) - \mathcal{F}'(g)(\delta)| = \left| \int_{[0,1]^d} \delta(\mathbf{x})^2 n^{-1} \sum_{i=1}^n \prod_{j=1}^d K_{h_j}(x_j, X_{ij}) d\mathbf{x} \right| = \|\delta\|_{2,n}^2 = o(\|\delta\|_{2,n})$$

as  $\|\delta\|_{2,n} \rightarrow 0$

## Alternative derivation of SBF equation

**Lemma 15.1:** Assume that the basic assumptions hold. Then, the equation

$$\mathcal{F}'(\hat{m})(\delta) = 0, \quad \delta \in \mathcal{H}_+$$

is equivalent to the system of the SBF equations,

$$\hat{m}_j(x_j) = \hat{f}_j(x_j) + \sum_{k \neq j}^d \int_0^1 \hat{m}_k(x_k) \frac{\hat{p}_{j,k}(x_j, x_k)}{\hat{p}_j(x_j)} dx_k, \quad 1 \leq j \leq d$$

## Alternative derivation of SBF equation

**Proof.**

$$\mathcal{F}'(\hat{m})(\delta) = -\frac{2}{n} \int_{[0,1]^d} \sum_{i=1}^n (Y_i - \hat{m}_1(x_1) - \cdots - \hat{m}_d(x_d)) \delta(\mathbf{x}) \prod_{j=1}^d K_{h_j}(x_j, X_{ij}) d\mathbf{x} = 0, \delta \in \mathcal{H}_+$$

is equivalent to

$$\int_{[0,1]^d} n^{-1} \sum_{i=1}^n (Y_i - \hat{m}_1(x_1) - \cdots - \hat{m}_d(x_d)) \delta_j(\mathbf{x}_j) \prod_{j=1}^d K_{h_j}(x_j, X_{ij}) d\mathbf{x} = 0$$

for  $\delta_j \in \mathcal{H}_j$ ,  $1 \leq j \leq d$ .

## Alternative derivation of SBF equation

$$\int_{[0,1]^d} n^{-1} \sum_{i=1}^n (Y_i - \hat{m}_1(x_1) - \cdots - \hat{m}_d(x_d)) \delta_j(\mathbf{x}_j) \prod_{j=1}^d K_{h_j}(x_j, X_{ij}) d\mathbf{x} = 0$$
$$\int_0^1 \delta_j(x_j) g_n(x_j) dx_j = 0, \quad \delta_j \in \mathcal{H}_j, 1 \leq j \leq d.$$

where  $g_n(x_j) = \int_{[0,1]^{d-1}} n^{-1} \sum_{i=1}^n (Y_i - \hat{m}_1(x_1) - \cdots - \hat{m}_d(x_d)) \prod_{j=1}^d K_{h_j}(x_j, X_{ij}) d\mathbf{x}_{-j}$ .  
is equivalent to

$$\int_{[0,1]^{d-1}} n^{-1} \sum_{i=1}^n (Y_i - \hat{m}_1(x_1) - \cdots - \hat{m}_d(x_d)) \cdot \prod_{j=1}^d K_{h_j}(x_j, X_{ij}) d\mathbf{x}_{-j} = 0, \quad 1 \leq j \leq d.$$

$$\hat{p}_j(x_j) \cdot \hat{f}_j(x_j) - \hat{p}_j(x_j) \cdot \hat{m}_j(x_j) - \sum_{k=1, k \neq j}^d \int_0^1 \hat{m}_k(x_k) \cdot \hat{p}_{j,k}(x_j, x_k) dx_k = 0, \quad 1 \leq j \leq d.$$

is equivalent to SBF equations.

## SBF Algorithm

Let  $(\hat{m}_j^{[0]}, 1 \leq j \leq d)$  be an initial tuple of component estimators. One could simply take  $\hat{m}_j^{[0]} \equiv 0$ .

For  $r \geq 1$ , compute

$$\begin{aligned}\hat{m}_j^{[r]}(x_j) &= \hat{f}_j(x_j) - \sum_{k=1}^{j-1} \int_0^1 \hat{m}_k^{[r]} \cdot \frac{\hat{p}_{j,k}(x_j, x_k)}{\hat{p}_j(x_j)} dx_k \\ &\quad - \sum_{k=j+1}^d \int_0^1 \hat{m}_k^{[r-1]}(x_k) \cdot \frac{\hat{p}_{j,k}(x_j, x_k)}{\hat{p}_j(x_j)} dx_k, \quad 1 \leq j \leq d\end{aligned}$$

Stop the iteration when

$$\int_0^1 \left| \hat{m}^{[r]}(\mathbf{x}) - \hat{m}^{[r-1]}(\mathbf{x}) \right|^2 \hat{p}(\mathbf{x}) d\mathbf{x} = \|\hat{m}^{[r]} - \hat{m}^{[r-1]}\|_{2,n}^2$$

is small, where  $\hat{m}^{[r]}(\mathbf{x}) = \hat{m}_1^{[r]}(x_1) + \cdots + \hat{m}_d^{[r]}(x_d)$ .



## Projection operators for SBF

Let  $\hat{\pi}_j : L^2([0, 1]^d, \mu_n) \rightarrow \mathcal{H}_j$  be the projection operator defined by

$$\hat{\pi}_j(g)(\mathbf{x}) \equiv \hat{\pi}_j(g)(x_j) = \int_{[0,1]^{d-1}} g(\mathbf{x}) \cdot \frac{\hat{p}(\mathbf{x})}{\hat{p}_j(x_j)} d\mathbf{x}_{-j}.$$

We can check

$$\int_0^1 \left( \int_{[0,1]^{d-1}} g(\mathbf{x}) \cdot \frac{\hat{p}(\mathbf{x})}{\hat{p}_j(x_j)} d\mathbf{x}_{-j} \right)^2 \hat{p}_j(x_j) dx_j \leq \int_0^1 \left( \int g(\mathbf{x})^2 \frac{\hat{p}(\mathbf{x})}{\hat{p}_j(x_j)} d\mathbf{x}_{-j} \int \frac{\hat{p}(\mathbf{x})}{\hat{p}_j(x_j)} d\mathbf{x}_{-j} \right) \hat{p}_j(x_j) dx_j$$

Then, the SBF equation

$$\hat{m}_j(x_j) = \hat{f}_j(x_j) + \sum_{k=1, k \neq j}^d \int_0^1 \hat{m}_k(x_k) \frac{\hat{p}_{j,k}(x_j, x_k)}{\hat{p}_j(x_j)} dx_k, \quad 1 \leq j \leq d$$

can be written as

$$\hat{m}_j = \hat{f}_j - \sum_{k=1, k \neq j}^d \hat{\pi}_j(\hat{m}_k), \quad 1 \leq j \leq d.$$

since  $\hat{m}_k \in L^2([0, 1]^d, \mu_n)$ . Also, using  $\hat{m}_j = \hat{\pi}_j(\hat{m}_j)$ , we get  $\hat{\pi}_j(\hat{m}) = \hat{f}_j$ .

## Projection operators for SBF

Recall  $\hat{f}_j(x_j) = \hat{p}_j(x_j)^{-1} \cdot n^{-1} \sum_{i=1}^n K_{h_j}(x_j, X_{ij}) Y_i \in \mathcal{H}_j$ .  $\hat{f}_j \in \mathcal{H}_j$  since

$$\begin{aligned} \int_0^1 \hat{f}_j(x_j)^2 \hat{p}_j(x_j) dx_j &= \int_0^1 \frac{(n^{-1} \sum_{i=1}^n K_{h_j}(x_j, X_{ij}) Y_i)^2}{\hat{p}_j(x_j)} dx_j \\ &\leq \int_0^1 \max_{1 \leq i \leq n} |Y_i| \cdot \hat{p}_j(x_j) dx_j = \max_{1 \leq i \leq n} |Y_i| < \infty \end{aligned}$$

for  $1 \leq j \leq d$ . Define  $\hat{f}_{\oplus}$  by

$$\hat{f}_{\oplus} = \hat{f}_d + (I - \hat{\pi}_d) \hat{f}_{d-1} + (I - \hat{\pi}_d) \circ (I - \hat{\pi}_{d-1}) \hat{f}_{d-2} + \cdots + (I - \hat{\pi}_d) \circ \cdots \circ (I - \hat{\pi}_2) \hat{f}_1.$$

$\hat{f}_{\oplus} \in \mathcal{H}_+$  since  $\int (f + g)^2 d\mu_n \leq 2(\int f^2 d\mu_n + \int g^2 d\mu_n) < \infty$ . (L2 space is a vector space)

## Projection operators for SBF

Also, define a *linear operator*  $\hat{\mathcal{T}} : \mathcal{H}_+ \rightarrow \mathcal{H}_+$  by

$$\hat{\mathcal{T}} = (I - \hat{\pi}_d) \circ \cdots \circ (I - \hat{\pi}_2) \circ (I - \hat{\pi}_1).$$

From  $\hat{\pi}_j(\hat{m}) = \hat{f}_j$ , we get

$$\hat{m} = \hat{f}_j + (I - \hat{\pi}_j)(\hat{m}), \quad 1 \leq j \leq d. \quad (*)$$

Applying (\*) successively for  $j = d$  to  $j = 1$ , we find

$$\begin{aligned} \hat{m} &= \hat{f}_d + (I - \hat{\pi}_d)(\hat{m}) \\ &= \hat{f}_d + (I - \hat{\pi}_d)\hat{f}_{d-1} + (I - \hat{\pi}_d) \circ (I - \hat{\pi}_{d-1})(\hat{m}) \\ &\quad \dots \\ &= \hat{f}_{\oplus} + \hat{\mathcal{T}}(\hat{m}). \end{aligned}$$

# Projection operators for SBF

## SBF Algorithm

$$\begin{aligned}\hat{m}_j^{[r]}(x_j) &= \hat{f}_j(x_j) - \sum_{k=1}^{j-1} \int_0^1 \hat{m}_k^{[r]} \cdot \frac{\hat{p}_{j,k}(x_j, x_k)}{\hat{p}_j(x_j)} dx_k \\ &\quad - \sum_{k=j+1}^d \int_0^1 \hat{m}_k^{[r-1]}(x_k) \cdot \frac{\hat{p}_{j,k}(x_j, x_k)}{\hat{p}_j(x_j)} dx_k, \quad 1 \leq j \leq d\end{aligned}$$

can be written as

$$\hat{m}_j^{[r]} = \hat{\pi}_j \left( \hat{m} - \sum_{k=1}^{j-1} \hat{m}_k^{[r]} - \sum_{k=j+1}^d \hat{m}_k^{[r-1]} \right)$$

## Projection operators for SBF

$$\begin{aligned}
 \hat{m} - \hat{m}^{[r]} &= \left[ \hat{m} - \hat{m}_1^{[r]} - \dots - \hat{m}_{d-1}^{[r]} \right] - \hat{m}_d^{[r]} = (I - \hat{\pi}_d) \left( \hat{m} - \hat{m}_1^{[r]} - \dots - \hat{m}_{d-1}^{[r]} \right) \\
 &= (I - \hat{\pi}_d) \left( \hat{m} - \hat{m}_1^{[r]} - \dots - \hat{m}_{d-2}^{[r]} - \hat{m}_d^{[r-1]} - \hat{m}_{d-1}^{[r]} \right) \\
 &= (I - \hat{\pi}_d)(I - \hat{\pi}_{d-1}) \left( \hat{m} - \hat{m}_1^{[r]} - \dots - \hat{m}_{d-2}^{[r]} - \hat{m}_d^{[r-1]} \right) \\
 &= (I - \hat{\pi}_d)(I - \hat{\pi}_{d-1}) \left( \hat{m} - \hat{m}_1^{[r]} - \dots - \hat{m}_{d-3}^{[r]} - \hat{m}_{d-1}^{[r-1]} - \hat{m}_d^{[r-1]} - \hat{m}_{d-2}^{[r]} \right) \\
 &\quad \vdots \\
 &= (I - \hat{\pi}_d) \cdots (I - \hat{\pi}_1) \left( \hat{m} - \hat{m}_1^{[r-1]} - \dots - \hat{m}_{d-1}^{[r-1]} - \hat{m}_d^{[r-1]} \right) = \hat{\mathcal{T}} \left( \hat{m} - \hat{m}^{[r-1]} \right)
 \end{aligned}$$

Now using  $\hat{f}_{\oplus} = (I - \hat{\mathcal{T}})(\hat{m})$ , we get

$$\hat{m}^{[r]} = \hat{f}_{\oplus} + \hat{\mathcal{T}}(\hat{m}^{[r-1]}), \quad r \geq 1.$$

## Existence of SBF estimator

Let the space  $L^2([0, 1]^d)$  is endowed with the norm  $\|\cdot\|_{2,n}$  defined by

$$\|g\|_{2,n}^2 = \int_{[0,1]^d} g(x)^2 \hat{p}(x) dx.$$

For  $\hat{T}$ , we define the operator norm  $\|\cdot\|_{\text{op},n}$  by

$$\|\hat{T}\|_{\text{op},n} = \sup\{ \|\hat{T}(g)\|_{2,n} : g \in \mathcal{H}_+, \|g\|_{2,n} = 1 \}.$$

The existences of the minimizer of the objective functional  $\mathcal{F}$  and of the solution of the SBF equation (15.4), respectively, rely on the conditions that  $\mathcal{H}_+$  is a *closed* subspace of  $L^2([0, 1]^d)$  and that the linear operator  $\hat{T}$  is a *contraction*, i.e.  $\|\hat{T}\|_{\text{op},n} < 1$ . In fact, the two conditions are equivalent.

## Closedness of $\mathcal{H}_+$ and contraction of $\mathcal{T}$

### Lemma 15.2 (Theorem 4.6, Xu and Zikatanov, 2002)

Let  $\mathcal{H}$  be a Hilbert space and  $\mathcal{H}_j$  for  $1 \leq j \leq d$  be subspaces of  $\mathcal{H}$ . Define  $\mathcal{H}_+ = \{g_1 + \cdots + g_d : g_j \in \mathcal{H}_j, 1 \leq j \leq d\}$ . Let  $\mathcal{P}_j : \mathcal{H} \rightarrow \mathcal{H}_j$  be orthogonal projections. Then, the followings are equivalent: (i)  $\mathcal{H}_+$  is a closed subspace of  $\mathcal{H}$ ; (ii)  $\|(I - \mathcal{P}_d) \circ \cdots \circ (I - \mathcal{P}_1)\| < 1$ .

### Lemma 15.3

Assume that the basic assumptions at (15.3) hold and that, for all  $1 \leq j \neq k \leq d$ ,

$$\int_{[0,1]^2} \frac{\widehat{p}_{j,k}(x_j, x_k)^2}{\widehat{p}_j(x_j) \widehat{p}_k(x_k)} dx_j dx_k < \infty. \quad (15.13)$$

Then,  $\mathcal{H}_+$  is closed and  $\|\widehat{T}\|_{\text{op},n} < 1$ .

## Closedness of $\mathcal{H}_+$ and contraction of $\mathcal{T}$

**Proof of Lemma 15.3.** We use Lemma 15.2 with  $\mathcal{H} = L^2([0, 1]^d)$ ,  $\mathcal{H}_+ = \mathcal{H}_+$  and  $\mathcal{H}_j = \mathcal{H}_j$ . We prove that  $\mathcal{H}_+$  is closed. For  $j \neq k$ , put

$$\hat{\kappa}_{j,k}(\mathbf{x}, \mathbf{u}) = \frac{\hat{p}_{j,k}(x_j, u_k)}{\hat{p}_j(x_j)\hat{p}_k(u_k)}.$$

Then, for  $g_k \in \mathcal{H}_k$ ,

$$\hat{\pi}_j(g_k)(\mathbf{x}) = \int_{[0,1]^d} \hat{\kappa}_{j,k}(\mathbf{x}, \mathbf{u}) \cdot g_k(\mathbf{u}) \cdot \hat{p}(\mathbf{u}) d\mathbf{u},$$

so that  $\hat{\kappa}_{j,k}$  is the ‘kernel’ of the linear operator  $\hat{\pi}_j$  restricted to  $\mathcal{H}_k$ .



## Is $L^2([0, 1]^d)$ a Hilbert space? (1/4)

**Setup.**

$$\mathcal{H} = L^2([0, 1]^d), \quad \|g\|_{2,n}^2 := \int_{[0,1]^d} g(\mathbf{x})^2 \hat{p}(\mathbf{x}) d\mathbf{x}.$$

**Inner product.**  $\langle f, g \rangle_{2,n} = \int_{[0,1]^d} f(\mathbf{x}) g(\mathbf{x}) \hat{p}(\mathbf{x}) d\mathbf{x}$ , which satisfies inner-product definition.

**Goal.** Given a Cauchy sequence  $(f_n) \subset \mathcal{H}$ , find  $f \in \mathcal{H}$  such that  $\|f_n - f\|_{2,n} \rightarrow 0$ .

**Fast Cauchy subsequence.** Choose indices  $n_0 = 0 < n_1 < n_2 < \dots$  so that

$$\|f_{n_{k+1}} - f_{n_k}\|_{2,n} \leq 2^{-k}, \quad k \geq 0,$$

and set  $f_{n_0} \equiv 0$ .

## Is $L^2([0, 1]^d)$ a Hilbert space? (2/4)

**Telescoping representation.** For each  $k \geq 1$ ,  $f_{n_{k+1}} = \sum_{j=0}^k (f_{n_{j+1}} - f_{n_j})$ .

**Square-summability.**

$$\left\| \sum_{j=0}^{\infty} |f_{n_{j+1}} - f_{n_j}| \right\|_{2,n} \leq \sum_{j=0}^{\infty} \|f_{n_{j+1}} - f_{n_j}\|_{2,n} = \|f_{n_1}\|_{2,n} + \sum_{j=1}^{\infty} 2^{-j} < \infty. \quad (*)$$

**Almost-sure convergence.**  $(*)$  implies  $\sum_{j=0}^{\infty} |f_{n_{j+1}}(\mathbf{x}) - f_{n_j}(\mathbf{x})| < \infty$  for  $\hat{p}$ -a.e.  $\mathbf{x}$ .

**Define**

$$f(\mathbf{x}) := \sum_{j=0}^{\infty} (f_{n_{j+1}}(\mathbf{x}) - f_{n_j}(\mathbf{x})).$$

Is  $L^2([0, 1]^d)$  a Hilbert space? (3/4)

**Bound  $\|f\|_{2,n}$ .** By Minkowski and (\*) we have  $\|f\|_{2,n} < \infty$ , so  $f \in \mathcal{H}$ .

**Distance to the subsequence.**

$$\|f - f_{n_k}\|_{2,n} = \left\| \sum_{j=k}^{\infty} (f_{n_{j+1}} - f_{n_j}) \right\|_{2,n} \leq \sum_{j=k}^{\infty} \|f_{n_{j+1}} - f_{n_j}\|_{2,n} \xrightarrow{k \rightarrow \infty} 0.$$

**Cauchy “tail” argument.** Given  $\varepsilon > 0$  choose  $N$  such that  $\|f_j - f_{n_k}\|_{2,n} < \varepsilon/2$  for all  $j, k \geq N$ . Fix  $k \geq N$ ; then for every  $j \geq N$ ,

$$\|f - f_j\|_{2,n} \leq \|f - f_{n_k}\|_{2,n} + \|f_{n_k} - f_j\|_{2,n} < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Is  $L^2([0, 1]^d)$  a Hilbert space? (4/4)

**Completeness.** We have exhibited  $f \in \mathcal{H}$  such that  $\|f_n - f\|_{2,n} \rightarrow 0$ . Hence  $(\mathcal{H}, \langle \cdot, \cdot \rangle_{2,n})$  is complete.

### Conclusion

$L^2([0, 1]^d)$ , equipped with the weighted norm  $\|\cdot\|_{2,n}$  and inner product  $\langle \cdot, \cdot \rangle_{2,n}$ , is a Hilbert space.  $\square$

## Closedness of $\mathcal{H}_+$ and contraction of $\mathcal{T}$

All the kernels  $\hat{\kappa}_{j,k}$  for  $j \neq k$  are **Hilbert–Schmidt** since

$$\int_{[0,1]^d} \int_{[0,1]^d} \hat{\kappa}_{j,k}(\mathbf{x}, \mathbf{u})^2 \cdot \hat{p}(\mathbf{u}) \hat{p}(\mathbf{x}) d\mathbf{u} d\mathbf{x} = \int_{[0,1]^2} \frac{\hat{p}_{j,k}(x_j, u_k)^2}{\hat{p}_j(x_j) \hat{p}_k(u_k)} dx_j du_k < \infty$$

by the marginalization properties at (15.7) and the condition (15.13).

Thus, all projections  $\hat{\pi}_j$  restricted to  $\mathcal{H}_k$  for  $1 \leq j \neq k \leq d$  are **compact** operators. According to Proposition A.4.2 in Bickel et al. (1993), this implies that  $\mathcal{H}_+$  is closed.  $\square$

## Detail for the projection $\hat{\pi}_j(g_k)$ in Lemma 15.3

$$\begin{aligned}\hat{\pi}_j(g_k)(x_j) &= \int_{[0,1]^d} g_k(\mathbf{u}) \frac{\hat{p}(\mathbf{u})}{\hat{p}_j(x_j)} d\mathbf{u}_{-j} \\ &= \int_{[0,1]} g_k(u_k) \frac{\hat{p}_{j,k}(x_j, u_k)}{\hat{p}_j(x_j)} du_k \\ &= \int_{[0,1]} g_k(u_k) \frac{\hat{p}_{j,k}(x_j, u_k)}{\hat{p}_j(x_j) \hat{p}_k(u_k)} \hat{p}_k(u_k) du_k \\ &= \int_{[0,1]^d} \hat{\kappa}_{j,k}(\mathbf{x}, \mathbf{u}) g_k(\mathbf{u}) \hat{p}(\mathbf{u}) d\mathbf{u} \quad \left( \hat{\kappa}_{j,k}(\mathbf{x}, \mathbf{u}) = \frac{\hat{p}_{j,k}(x_j, u_k)}{\hat{p}_j(x_j) \hat{p}_k(u_k)} \right).\end{aligned}$$

Thus  $\hat{\kappa}_{j,k}$  is precisely the kernel of the projection  $\hat{\pi}_j$  when it is restricted to the subspace  $\mathcal{H}_k$ .

## Is $\hat{\pi}_j$ an orthogonal projection?

Let  $g \in L^2([0, 1]^d)$  and  $h \in \mathcal{H}_j$  (i.e.  $h(\mathbf{x}) = h_j(x_j)$ ). To verify orthogonality we must show  $\langle g - \hat{\pi}_j(g), h \rangle_{2,n} = 0$ , equivalently  $\langle g, h \rangle_{2,n} = \langle \hat{\pi}_j(g), h \rangle_{2,n}$ .

$$\begin{aligned}\langle g, h \rangle_{2,n} &= \int_{[0,1]^d} g(\mathbf{x}) h_j(x_j) \hat{p}(\mathbf{x}) d\mathbf{x} \\ &= \int_0^1 \left[ \int_{[0,1]^{d-1}} g(\mathbf{x}) \frac{\hat{p}(\mathbf{x})}{\hat{p}_j(x_j)} d\mathbf{x}_{-j} \right] \hat{p}_j(x_j) h_j(x_j) dx_j \\ &= \int_0^1 \hat{\pi}_j(g)(x_j) \hat{p}_j(x_j) h_j(x_j) dx_j = \langle \hat{\pi}_j(g), h \rangle_{2,n},\end{aligned}$$

where the second line uses Fubini's theorem and the definition of the marginal density  $\hat{p}_j$ ; the third line recognises the inner integral as  $\hat{\pi}_j(g)(x_j)$ .

Hence  $\langle g - \hat{\pi}_j(g), h \rangle_{2,n} = 0$  for all  $h \in \mathcal{H}_j$ , so  $\hat{\pi}_j$  is an *orthogonal projection* onto  $\mathcal{H}_j$ .

# Integral form of the restricted projection

**Integral linear operators.** For a kernel  $K(\mathbf{x}, \mathbf{y})$  the map  $(Tf)(\mathbf{x}) = \int_{[0,1]^d} f(\mathbf{y})K(\mathbf{x}, \mathbf{y}) d\mathbf{y}$  defines a linear operator on  $L^2([0, 1]^d)$ .

**Our setting.**

$$\hat{\pi}_j(g_k)(\mathbf{x}) = \int_{[0,1]^d} \hat{\kappa}_{j,k}(\mathbf{x}, \mathbf{u}) g_k(\mathbf{u}) \hat{p}(\mathbf{u}) d\mathbf{u}, \quad g_k \in \mathcal{H}_k,$$

with

$$\hat{\kappa}_{j,k}(\mathbf{x}, \mathbf{u}) = \frac{\hat{p}_{j,k}(x_j, u_k)}{\hat{p}_j(x_j)\hat{p}_k(u_k)}.$$

Thus the projection  $\hat{\pi}_j$  restricted to  $\mathcal{H}_k$  is an integral operator on the Hilbert space  $\mathcal{H} = L^2([0, 1]^d, \hat{p})$ .



## Finite Hilbert–Schmidt norm of $\hat{\kappa}_{j,k}$

$$\begin{aligned} & \int_{[0,1]^d} \int_{[0,1]^d} \hat{\kappa}_{j,k}(\mathbf{x}, \mathbf{u})^2 \hat{p}(\mathbf{x}) \hat{p}(\mathbf{u}) d\mathbf{u} d\mathbf{x} \\ &= \int_{[0,1]^d} \hat{p}(\mathbf{x}) \int_{[0,1]} \hat{\kappa}_{j,k}(\mathbf{x}, \mathbf{u})^2 \hat{p}_k(u_k) du_k d\mathbf{x} \\ &= \int_{[0,1]} \hat{p}_k(u_k) \int_{[0,1]^d} \hat{\kappa}_{j,k}(\mathbf{x}, \mathbf{u})^2 \hat{p}(\mathbf{x}) d\mathbf{x} du_k \\ &= \int_{[0,1]^2} \frac{\hat{p}_{j,k}(x_j, u_k)^2}{\hat{p}_j(x_j) \hat{p}_k(u_k)} dx_j du_k. \end{aligned}$$

By assumption  $\int_{[0,1]^2} \frac{\hat{p}_{j,k}(x_j, u_k)^2}{\hat{p}_j(x_j) \hat{p}_k(u_k)} dx_j du_k < \infty$ , so the kernel is Hilbert–Schmidt.

**Consequence.** An operator with Hilbert–Schmidt kernel is compact. Hence each restricted projection  $\hat{\pi}_j|_{\mathcal{H}_k}$  is compact. By Proposition A.4.2 of Bickel et al. (1993) this compactness implies that the sum space  $\mathcal{H}_+ = \sum_{j=1}^d \mathcal{H}_j$  is closed in  $\mathcal{H}$ . □

## Existence of SBF estimator

**Theorem 15.1 (Non-asymptotic existence)** Assume the conditions of Lemma 15.3. Then, there exists a unique solution  $\hat{m} \in \mathcal{H}_+$  of the system of equations at (15.4).  $\square$

The proof of Theorem 15.1 relies on the following two lemmas. Let  $\mathcal{H}$  be a Hilbert space equipped with a norm  $\|\cdot\|$ .

**Lemma 15.4 (Lemma 4, Beltrami, 1967)** Let  $\mathcal{H}_0$  be a closed convex subset of  $\mathcal{H}$ . Assume that  $f : \mathcal{H}_0 \rightarrow \mathbb{R}$  is (strictly) convex, continuous and  $f(x) \rightarrow +\infty$  as  $\|x\| \rightarrow \infty$ . Then,  $f$  attains a (unique) minimum in  $\mathcal{H}_0$ .  $\square$

## Existence of SBF estimator

**Lemma 15.5 (Theorem 5.3.19, Atkinson & Han, 2009)** Let  $\mathcal{H}_0$  be a nonempty convex subspace of  $\mathcal{H}$ . Assume that  $f : \mathcal{H}_0 \rightarrow \mathbb{R}$  is convex and Gâteaux differentiable. Then,  $f$  has a minimum  $x_0$  in  $\mathcal{H}_0$  if and only if  $f'(x_0) = 0$ .  $\square$

**Proof of Theorem 15.1.** First, note that  $\mathcal{F}$  is strictly convex on  $\mathcal{H}_+$ . Also, it is continuous since

$$|\mathcal{F}(f) - \mathcal{F}(g)| \leq \|f - g\|_{2,n} \left( 2 \max_{1 \leq i \leq n} |Y_i| + 2\|f\|_{2,n} + \|f - g\|_{2,n} \right).$$

## Strict convexity of $\mathcal{F}$ (1/2)

For  $g \in \mathcal{H}_+$  write

$$\mathcal{F}(g) = \frac{1}{n} \sum_{i=1}^n \int_{[0,1]^d} \{Y_i - g_1(x_1) - \cdots - g_d(x_d)\}^2 \prod_{j=1}^d K_{h_j}(x_j, X_{ij}) d\mathbf{x}.$$

**Definition of strict convexity.**  $\mathcal{F}$  is strictly convex iff for all  $0 < \lambda < 1$  and  $f, g \in \mathcal{H}_+$  with  $f \neq g$ ,

$$\mathcal{F}(\lambda g + (1 - \lambda)f) < \lambda \mathcal{F}(g) + (1 - \lambda) \mathcal{F}(f).$$

**Reduce to the integrand.** Since the outer sum and integral use non-negative weights, it suffices to show, for every  $i$  and  $\mathbf{x}$ ,

$$(Y_i - \lambda G - (1 - \lambda)F)^2 < \lambda (Y_i - G)^2 + (1 - \lambda) (Y_i - F)^2,$$

where

$$G = g_1(x_1) + \cdots + g_d(x_d), \quad F = f_1(x_1) + \cdots + f_d(x_d).$$

## Strict convexity of $\mathcal{F}$ (2/2)

**Pointwise comparison of squares.** Expanding the left-hand side (LHS) and right-hand side (RHS):

$$\text{LHS} = Y_i^2 - 2Y_i(\lambda G + (1 - \lambda)F) + (\lambda G + (1 - \lambda)F)^2,$$

$$\text{RHS} = Y_i^2 - 2Y_i(\lambda G + (1 - \lambda)F) + \lambda G^2 + (1 - \lambda)F^2.$$

Thus the difference RHS – LHS equals

$$\lambda G^2 + (1 - \lambda)F^2 - (\lambda G + (1 - \lambda)F)^2 = \lambda(1 - \lambda)(G - F)^2 > 0,$$

The strict inequality follows from the strict convexity of  $x \mapsto x^2$ .

**Conclusion.** The pointwise strict inequality propagates through the integral and the average over  $i$ , giving

$$\mathcal{F}(\lambda g + (1 - \lambda)f) < \lambda \mathcal{F}(g) + (1 - \lambda) \mathcal{F}(f),$$

so  $\mathcal{F}$  is *strictly convex*.

## Continuity of $\mathcal{F}$

Let  $f, g \in \mathcal{H}_+$  and set  $f(\mathbf{x}) = \sum_{j=1}^d f_j(x_j)$ ,  $g(\mathbf{x}) = \sum_{j=1}^d g_j(x_j)$ . Then

$$\begin{aligned}\mathcal{F}(f) - \mathcal{F}(g) &= \frac{1}{n} \sum_{i=1}^n \int_{[0,1]^d} \left[ \{Y_i - f(\mathbf{x})\}^2 - \{Y_i - g(\mathbf{x})\}^2 \right] \prod_{j=1}^d K_{h_j}(x_j, X_{ij}) d\mathbf{x} \\ &= \frac{1}{n} \sum_{i=1}^n \int_{[0,1]^d} (f(\mathbf{x}) - g(\mathbf{x})) (f(\mathbf{x}) + g(\mathbf{x}) - 2Y_i) \prod_{j=1}^d K_{h_j}(x_j, X_{ij}) d\mathbf{x}.\end{aligned}$$

## Continuity of $\mathcal{F}$

**Taking absolute values.**

$$\begin{aligned} |\mathcal{F}(f) - \mathcal{F}(g)| &\leq \frac{1}{n} \sum_{i=1}^n \int_{[0,1]^d} |f(\mathbf{x}) - g(\mathbf{x})| \left( |f(\mathbf{x})| + |g(\mathbf{x})| + 2 \max_{1 \leq i \leq n} |Y_i| \right) \prod_{j=1}^d K_{h_j}(x_j, X_{ij}) d\mathbf{x} \\ &= \int_{[0,1]^d} |f(\mathbf{x}) - g(\mathbf{x})| \left( |f(\mathbf{x})| + |g(\mathbf{x})| + 2 \max_{1 \leq i \leq n} |Y_i| \right) \hat{p}(\mathbf{x}) d\mathbf{x} \end{aligned} \quad (*)$$

**Bounding the integrand.** Because  $|g(\mathbf{x})| \leq |f(\mathbf{x}) - g(\mathbf{x})| + |f(\mathbf{x})|$ , substituting into (\*) and applying Cauchy–Schwarz yields

$$|\mathcal{F}(f) - \mathcal{F}(g)| \leq \|f - g\|_{2,n} \left( 2 \max_{1 \leq i \leq n} |Y_i| + 2\|f\|_{2,n} + \|f - g\|_{2,n} \right),$$

establishing the required continuity estimate.

## Existence of SBF estimator

Furthermore, it holds that

$$\mathcal{F}(g) \geq \min_{1 \leq i \leq n} |Y_i|^2 + \|g\|_{2,n}^2 \left( 1 - \frac{2 \max_{1 \leq i \leq n} |Y_i|}{\|g\|_{2,n}} \right),$$

so that  $\mathcal{F}(g)$  diverges to infinity as  $\|g\|_{2,n} \rightarrow \infty$ . By Lemma 15.3,  $\mathcal{H}_+$  is closed. Thus, according to Lemma 15.4 there exists a unique minimizer of  $\mathcal{F}$  in  $\mathcal{H}_+$ .

Now, according to Lemma 15.5, the minimization of  $\mathcal{F}(g)$  over  $g \in \mathcal{H}_+$  is equivalent to solving  $\mathcal{F}'(g) = 0$  for  $g \in \mathcal{H}_+$ . By Lemma 15.1, the latter is also equivalent to solving the system of equations at (15.4).

This implies that (15.4) has a unique solution in  $\mathcal{H}_+$ . □



## Divergence of $\mathcal{F}(g)$ (1/2)

$$\begin{aligned}\mathcal{F}(g) &= \frac{1}{n} \sum_{i=1}^n \int_{[0,1]^d} \{Y_i - g(\mathbf{x})\}^2 \prod_{j=1}^d K_{h_j}(x_j, X_{ij}) d\mathbf{x} \\ &= \frac{1}{n} \sum_{i=1}^n \int_{[0,1]^d} \left( Y_i^2 - 2g(\mathbf{x})Y_i + g(\mathbf{x})^2 \right) \prod_{j=1}^d K_{h_j}(x_j, X_{ij}) d\mathbf{x}.\end{aligned}$$

Because  $\min_{1 \leq i \leq n} |Y_i|^2 \leq Y_i^2$  and  $-2g(\mathbf{x})Y_i \geq -2g(\mathbf{x}) \max_{1 \leq i \leq n} |Y_i|$ , we obtain

$$\mathcal{F}(g) \geq \int_{[0,1]^d} \left( \min_{1 \leq i \leq n} |Y_i|^2 - 2g(\mathbf{x}) \max_{1 \leq i \leq n} |Y_i| + g(\mathbf{x})^2 \right) \hat{p}(\mathbf{x}) d\mathbf{x}.$$

## Divergence of $\mathcal{F}(g)$ (2/2)

Rewrite the bound:

$$\mathcal{F}(g) \geq \min_{1 \leq i \leq n} |Y_i|^2 - 2 \max_{1 \leq i \leq n} |Y_i| \int_{[0,1]^d} g(\mathbf{x}) \hat{p}(\mathbf{x}) d\mathbf{x} + \|g\|_{2,n}^2.$$

**Cauchy–Schwarz inequality.**

$$\left| \int_{[0,1]^d} g(\mathbf{x}) \hat{p}(\mathbf{x}) d\mathbf{x} \right| \leq \|g\|_{2,n},$$

hence

$$\boxed{\mathcal{F}(g) \geq \min_{1 \leq i \leq n} |Y_i|^2 + \|g\|_{2,n}^2 \left( 1 - \frac{2 \max_{1 \leq i \leq n} |Y_i|}{\|g\|_{2,n}} \right)}.$$

**Conclusion.** As  $\|g\|_{2,n} \rightarrow \infty$ , the bracketed term  $(1 - 2 \max |Y_i| / \|g\|_{2,n})$  tends to 1, so  $\mathcal{F}(g) \rightarrow \infty$ .

# Convergence of SBF algorithm

## Theorem 15.2 (Non-asymptotic convergence)

Assume the conditions of Lemma 15.3. Then, it holds that

$$\|\hat{m}^{[r]} - \hat{m}\|_{2,n} \leq \left( \frac{\|\hat{f}_{\oplus}\|_{2,n}}{1 - \|\hat{\mathcal{T}}\|_{\text{op},n}} + \|\hat{m}^{[0]}\|_{2,n} \right) \cdot \|\hat{\mathcal{T}}\|_{\text{op},n}^r, \quad r \geq 1.$$

**Note 15.2.** Since  $\|\hat{\mathcal{T}}\|_{\text{op},n} < 1$ , Theorem 15.2 implies that the  $r$ th update  $\hat{m}^{[r]}$  of the SBF algorithm (15.5) or equivalently (15.12) converges to the solution  $\hat{m}$  of the SBF equation (15.4). □

**Proof.** From Lemma 15.3,  $\|\hat{\mathcal{T}}\|_{\text{op},n} < 1$  and  $\mathcal{H}_+$  is closed. These and (15.12) imply that

# Convergence of the SBF algorithm

$$\hat{m}^{[\infty]} := \lim_{r \rightarrow \infty} \hat{m}^{[r]} = \lim_{r \rightarrow \infty} \left( \sum_{k=0}^{r-1} \hat{\mathcal{T}}^k(\hat{f}_{\oplus}) + \hat{\mathcal{T}}^r(\hat{m}^{[0]}) \right) = \sum_{k=0}^{\infty} \hat{\mathcal{T}}^k(\hat{f}_{\oplus})$$

exists in  $\mathcal{H}_+$ . Since

$$\hat{m}^{[\infty]} = \hat{f}_{\oplus} + \hat{\mathcal{T}}(\hat{m}^{[\infty]}), \quad \text{and Theorem 15.1 guarantees uniqueness,}$$

$$\boxed{\hat{m} = \hat{m}^{[\infty]}}$$

$$\begin{aligned} \|\hat{m}^{[r]} - \hat{m}\|_{2,n} &= \left\| - \sum_{k=r}^{\infty} \hat{\mathcal{T}}^k(\hat{f}_{\oplus}) + \hat{\mathcal{T}}^r(\hat{m}^{[0]}) \right\|_{2,n} \\ &\leq \sum_{k=r}^{\infty} \|\hat{\mathcal{T}}\|_{\text{op},n}^k \|\hat{f}_{\oplus}\|_{2,n} + \|\hat{\mathcal{T}}\|_{\text{op},n}^r \|\hat{m}^{[0]}\|_{2,n} \\ &= \frac{\|\hat{\mathcal{T}}\|_{\text{op},n}^r}{1 - \|\hat{\mathcal{T}}\|_{\text{op},n}} \|\hat{f}_{\oplus}\|_{2,n} + \|\hat{\mathcal{T}}\|_{\text{op},n}^r \|\hat{m}^{[0]}\|_{2,n}. \end{aligned}$$

This completes the proof. □

## Details for Theorem 15.2 (I)

**Iteration (15.12).**

$$\hat{m}^{[r]} = \hat{f}_{\oplus} + \hat{\mathcal{T}}(\hat{m}^{[r-1]}), \quad r \geq 1.$$

**Limit of the sequence.** Using the previous page's inequalities and the fact that  $\|\hat{\mathcal{T}}\|_{\text{op},n} < 1$ ,

$$\lim_{r \rightarrow \infty} \hat{m}^{[r]} = \sum_{k=0}^{\infty} \hat{\mathcal{T}}^k(\hat{f}_{\oplus}) =: \hat{m}^{[\infty]} \in \mathcal{H}_+ \quad (\text{by closedness of } \mathcal{H}_+).$$

Because  $\hat{\mathcal{T}}$  is bounded (hence continuous), taking  $r \rightarrow \infty$  in the recursion yields

$$\hat{m}^{[\infty]} = \hat{f}_{\oplus} + \hat{\mathcal{T}}(\hat{m}^{[\infty]}).$$

**Uniqueness.** By the uniqueness of the solution  $\hat{m}$  established in Theorem 15.1, we obtain  $\hat{m} = \hat{m}^{[\infty]}$ .

## Details for Theorem 15.2 (II)

For  $r \geq 1$ ,

$$\begin{aligned}\|\hat{m}^{[r]} - \hat{m}\|_{2,n} &= \left\| -\sum_{k=r}^{\infty} \hat{\mathcal{T}}^k(\hat{f}_{\oplus}) + \hat{\mathcal{T}}^r(\hat{m}^{[0]}) \right\|_{2,n} \\ &\leq \sum_{k=r}^{\infty} \left\| \hat{\mathcal{T}}^k(\hat{f}_{\oplus}) \right\|_{2,n} + \left\| \hat{\mathcal{T}}^r(\hat{m}^{[0]}) \right\|_{2,n} \\ &\leq \sum_{k=r}^{\infty} \|\hat{\mathcal{T}}\|_{\text{op},n}^k \|\hat{f}_{\oplus}\|_{2,n} + \|\hat{\mathcal{T}}\|_{\text{op},n}^r \|\hat{m}^{[0]}\|_{2,n} \\ &= \frac{\|\hat{\mathcal{T}}\|_{\text{op},n}^r}{1 - \|\hat{\mathcal{T}}\|_{\text{op},n}} \|\hat{f}_{\oplus}\|_{2,n} + \|\hat{\mathcal{T}}\|_{\text{op},n}^r \|\hat{m}^{[0]}\|_{2,n}.\end{aligned}$$

(The third inequality uses the definition of the operator norm.)

## Existence and convergence: asymptotic properties

### Theorem 15.3 (Asymptotic versions of Theorems 15.1 & 15.2)

Assume that  $p_j$  for  $1 \leq j \leq d$  are bounded away from zero on  $[0, 1]$  and  $p_{j,k}$  for  $1 \leq j \neq k \leq d$  are continuous on  $[0, 1]^2$ . Suppose that  $K$  is Lipschitz continuous and the support of  $K$  contains an interval with non-empty interior on each of  $[-1, 0]$  and  $[0, 1]$ . Also, assume that  $h_j \rightarrow 0$  for  $1 \leq j \leq d$  and  $nh_j h_k / \log n \rightarrow \infty$  for  $1 \leq j \neq k \leq d$  as  $n \rightarrow \infty$ . Then, with probability tending to one, the system of equations at (15.4) has a unique solution and the  $r$ th update in the SBF algorithm at (15.5) converges to the solution of (15.4) as  $r \rightarrow \infty$ .  $\square$

## Existence and convergence: asymptotic properties

**Proof of Theorem 15.3.** In view of Lemma 15.3, it suffices to prove that there exist  $0 < \varepsilon, C < \infty$  such that, with probability tending to one,

$$\inf_{x_j \in [0,1]} \hat{p}_j(x_j) > \varepsilon, \quad 1 \leq j \leq d, \quad (15.14)$$

$$\int_{[0,1]^2} \frac{\hat{p}_{j,k}(x_j, x_k)^2}{\hat{p}_j(x_j) \hat{p}_k(x_k)} dx_j dx_k < C, \quad 1 \leq j \neq k \leq d. \quad (15.15)$$

First, we note that

$$K_h(u-v) \leq K_h(u, v) \leq \frac{K_h(u-v)}{\min\left\{\int_{-1}^0 K(u) du, \int_0^1 K(u) du\right\}}. \quad (15.16)$$

Put

$$\varepsilon_j = \frac{1}{2} \inf_{x_j \in [0,1]} p_j(x_j) \cdot \min\left\{\int_{-1}^0 K(u) du, \int_0^1 K(u) du\right\} > 0.$$



## Normalized kernel $K_h(x, u)$

**Definition.**

$$K_h(x, u) = \frac{K\left(\frac{x - u}{h}\right)}{\int_0^1 K\left(\frac{v - u}{h}\right) dv}.$$

**Basic bound.** Because

$$0 < \int_0^1 K\left(\frac{v - u}{h}\right) dv \leq 1,$$

we have

$$K_h(x - u) \leq K_h(x, u).$$

The remaining steps refine this comparison.

## Normalized kernel

$$\begin{aligned} K_h(x - u) &\leq K_h(x, u) \\ &= \frac{K\left(\frac{x - u}{h}\right)}{\int_0^1 K\left(\frac{v - u}{h}\right) dv} \\ &= \frac{K_h(x - u)}{\int_0^1 K\left(\frac{v - u}{h}\right) dv} \\ &\leq \frac{K_h(u - v)}{\min\left\{\int_{-1}^0 K(u) du, \int_0^1 K(u) du\right\}}. \end{aligned}$$

In the final step, the denominator is bounded below by the minimum of its two values on the sub-intervals  $(-1, 0)$  and  $(0, 1)$ , completing the inequality (15.16). □

## Existence and convergence: asymptotic properties

To prove (15.14), we use the first inequality of (15.16) to obtain

$$\begin{aligned}\inf_{x_j \in [0,1]} \hat{p}_j(x_j) &\geq \inf_{x_j \in [0,1]} n^{-1} \sum_{i=1}^n K_{h_j}(x_j - X_{ij}) \\ &\geq 2\varepsilon_j - \sup_{x_j \in [0,1]} \left| n^{-1} \sum_{i=1}^n K_{h_j}(x_j - X_{ij}) - \mathbb{E}\{K_{h_j}(x_j - X_j)\} \right| \\ &> \varepsilon_j.\end{aligned}$$

where the last inequality holds with probability tending to one. Taking

$$\varepsilon = \min_{1 \leq j \leq d} \varepsilon_j$$

completes the proof of (15.14).

## Detail for (15.14) – decomposition of $\hat{p}_j$

$$\hat{p}_j(x_j) = \frac{1}{n} \sum_{i=1}^n K_{h_j}(x_j, X_{ij}) \geq \frac{1}{n} \sum_{i=1}^n K_{h_j}(x_j - X_{ij}),$$

so that

$$\begin{aligned} \hat{p}_j(x_j) &\geq \frac{1}{n} \sum_{i=1}^n K_{h_j}(x_j - X_{ij}) - \mathbb{E}[K_{h_j}(x_j - X_j)] + \mathbb{E}[K_{h_j}(x_j - X_j)] \\ &= |\mathbb{E}[K_{h_j}(x_j - X_j)]| - \left| \frac{1}{n} \sum_{i=1}^n K_{h_j}(x_j - X_{ij}) - \mathbb{E}[K_{h_j}(x_j - X_j)] \right|. \end{aligned}$$

Taking the infimum over  $x_j \in [0, 1]$  gives

$$\inf_{x_j \in [0,1]} \hat{p}_j(x_j) \geq \inf_{x_j \in [0,1]} |\mathbb{E}[K_{h_j}(x_j - X_j)]| - \sup_{x_j \in [0,1]} \left| \frac{1}{n} \sum_{i=1}^n K_{h_j}(x_j - X_{ij}) - \mathbb{E}[K_{h_j}(x_j - X_j)] \right|.$$

## Detail for (15.14) – bounding the two terms

**First term.**

$$\begin{aligned}\mathbb{E}[K_{h_j}(x_j - X_j)] &= \int_0^1 K_{h_j}(x_j - v) p(v) dv \geq \left( \inf_{x_j \in [0,1]} p(x_j) \right) \int_0^1 K_{h_j}(x_j - v) dv \\ &\geq \left( \inf_{x_j \in [0,1]} p(x_j) \right) \min \left\{ \int_{-1}^0 K(u) du, \int_0^1 K(u) du \right\} = 2\varepsilon_j.\end{aligned}$$

**Second term.** By Lemma 6.2,

$$\sup_{x_j \in [0,1]} \left| n^{-1} \sum_{i=1}^n K_{h_j}(x_j - X_{ij}) - \mathbb{E}[K_{h_j}(x_j - X_j)] \right| = O_p(n^{-1/2} h_j^{-1/2} \sqrt{\log n}) = o_p(1).$$

With probability tending to one,

$$\inf_{x_j \in [0,1]} \hat{p}_j(x_j) > \varepsilon_j.$$

Finally set  $\varepsilon = \min_{1 \leq j \leq d} \varepsilon_j$ , yielding inequality (15.14). □

## Existence and convergence: asymptotic properties

Because of (15.14), to establish (15.15) it suffices to prove

$$\sup_{(x_j, x_k) \in [0,1]^2} \hat{p}_{j,k}(x_j, x_k) < C' \quad (1)$$

with probability tending to one for some constant  $0 < C' < \infty$ .

Using the second inequality of (15.16) we have

$$\sup_{(x_j, x_k) \in [0,1]^2} \mathbb{E}\{K_{h_j}(x_j - X_j)K_{h_k}(x_k - X_k)\} \leq \sup_{(x_j, x_k) \in [0,1]^2} p_{j,k}(x_j, x_k), \quad (2)$$

$$\sup_{(x_j, x_k) \in [0,1]^2} \left| n^{-1} \sum_{i=1}^n K_{h_j}(x_j - X_{ij})K_{h_k}(x_k - X_{ik}) - \mathbb{E}\{K_{h_j}(x_j - X_j)K_{h_k}(x_k - X_k)\} \right| = o_p(1). \quad (3)$$

(2) follows directly from  $\int_0^1 K_h(x - u) du \leq 1$ .

(3) is obtained analogously to Lemma 6.2 in *LecNote-Weeks 6*, using the condition  $nh_jh_k/\log n \rightarrow \infty$ .

□

## Details for (15.15)

**Goal.**

$$\sup_{(x_j, x_k) \in [0,1]^2} \hat{p}_{j,k}(x_j, x_k) < C' \quad (1)$$

with probability  $\rightarrow 1$ .

**Normalized  $\rightarrow$  unnormalized kernel.** Let

$$m := \min \left\{ \int_{-1}^0 K(u) du, \int_0^1 K(u) du \right\} > 0.$$

The second inequality of (15.16) gives  $K_h(u, v) \leq K_h(u - v)/m$ , so for any  $(x_j, x_k) \in [0, 1]^2$

$$\hat{p}_{j,k}(x_j, x_k) = \frac{1}{n} \sum_{i=1}^n K_{h_j}(x_j, X_{ij}) K_{h_k}(x_k, X_{ik}) \leq \frac{1}{m^2} \frac{1}{n} \sum_{i=1}^n K_{h_j}(x_j - X_{ij}) K_{h_k}(x_k - X_{ik}).$$

## Details for (15.15) – Bound (2))

**Bound (2).** For fixed  $(x_j, x_k) \in [0, 1]^2$

$$\begin{aligned} & \mathbb{E}\{K_{h_j}(x_j - X_j)K_{h_k}(x_k - X_k)\} \\ &= \int_{[0,1]^2} K_{h_j}(x_j - u_j) K_{h_k}(x_k - u_k) p_{j,k}(u_j, u_k) du_j du_k \\ &\leq \sup_{(u_j, u_k) \in [0,1]^2} p_{j,k}(u_j, u_k) < \infty \end{aligned}$$

Taking the supremum over  $(x_j, x_k)$  on the left-hand side yields exactly inequality (2). And finiteness is from the Maximum value theorem.



## Bound (3): definition and Lipschitz control

$$Z_n(x_j, x_k) := \frac{1}{n} \sum_{i=1}^n K_{h_j}(x_j - X_{ij}) K_{h_k}(x_k - X_{ik}) - \mathbb{E}\{K_{h_j}(x_j - X_j) K_{h_k}(x_k - X_k)\}.$$

**Lipschitz continuity of  $K(\cdot)K(\cdot)$ .**

$$\begin{aligned} & |K(y_1)K(y_2) - K(x_1)K(x_2)| \\ &= |K(y_1)K(y_2) - K(y_1)K(x_2) + K(y_1)K(x_2) - K(x_1)K(x_2)| \\ &\leq |K(y_1)| \cdot |K(y_2) - K(x_2)| + |K(x_2)| \cdot |K(y_1) - K(x_1)| \\ &\leq C'(|y_1 - x_1| + |y_2 - x_2|). \end{aligned}$$

$$\begin{aligned} |Z_n(x_j, x_k) - Z_n(x'_j, x'_k)| &\leq \frac{1}{n} \sum_{i=1}^n |K_{h_j}(x_j - X_{ij}) K_{h_k}(x_k - X_{ik}) - K_{h_j}(x'_j - X_{ij}) K_{h_k}(x'_k - X_{ik})| \\ &\leq \frac{C'}{n h_j h_k} \sum_{i=1}^n (|x_j - x'_j| + |x_k - x'_k|) = \frac{C'}{h_j h_k} (|x_j - x'_j| + |x_k - x'_k|). \end{aligned}$$

## Bound (3): $\delta$ -net argument

**Choose**  $\delta = n^{-\gamma}$  ( $0 < \gamma < 1$ ) and a  $\delta$ -net  $\mathcal{N}_\delta \subset [0, 1]$ . Then

$$\sup_{\substack{|x_j - x'_j| \leq \delta \\ |x_k - x'_k| \leq \delta}} |Z_n(x_j, x_k) - Z_n(x'_j, x'_k)| \leq \frac{C'}{h_j h_k} \delta^2 = \frac{2C'}{h_j h_k} n^{-2\gamma} \xrightarrow{\text{Lemma 6.2}} o_p\left((nh_j h_k)^{-1/2} \sqrt{\log n}\right).$$

**Variance bound for a single summand.**

$$\begin{aligned} \text{Var}(K_{h_j}(x_j - X_{ij})K_{h_k}(x_k - X_{ik})) &\leq \mathbb{E}[K_{h_j}^2(x_j - X_j)K_{h_k}^2(x_k - X_k)] \\ &\leq \frac{C}{h_j h_k}. \end{aligned}$$

**Define**  $\xi_{ni} = K_{h_j}(x_j - X_{ij})K_{h_k}(x_k - X_{ik}) - \mathbb{E}[K_{h_j}(x_j - X_j)K_{h_k}(x_k - X_k)]$  with  $\mathbb{E}(\xi_{ni}) = 0$  and  $\text{Var}(\xi_{ni}) \leq C/(h_j h_k)$ .

## Bound (3)

$$\begin{aligned}\text{Var}\left(K_{h_j}(x_j - X_{ij})K_{h_k}(x_k - X_{ik})\right) &= \mathbb{E}\left[K_{h_j}^2(x_j - X_j) K_{h_k}^2(x_k - X_k)\right] \\ &= \int_{[0,1]^2} K_{h_j}^2(x_j - u) K_{h_k}^2(x_k - v) p_{j,k}(u, v) du dv. \\ &= \int_{[0,1]^2} K_{h_j}^2(x_j - u) K_{h_k}^2(x_k - v) p_{j,k}(u, v) du dv \\ &= h_j^{-1} h_k^{-1} \int_0^1 \int_0^1 (K^2)_{h_j}(x_j - u) (K^2)_{h_k}(x_k - v) p_{j,k}(u, v) du dv.\end{aligned}$$

## Bound (3)

Using  $\sup_{(u,v) \in [0,1]^2} p_{j,k}(u,v) < \infty$  and Lipschitz Continuity of  $\int K^2(s) ds < \infty$ ,

$$\text{Var}\left(K_{h_j}(x_j - X_{ij})K_{h_k}(x_k - X_{ik})\right) \leq \frac{C}{h_j h_k}.$$

**Implication for the empirical sum.** With  $\xi_{ni}$  as defined on the previous slide ( $\mathbb{E}\xi_{ni} = 0$ ),

$$\mathbb{E}(\xi_{ni}^2) \leq \frac{C}{h_j h_k},$$

so by an inequality (as in Lemma 6.2) and the condition  $nh_j h_k / \log n \rightarrow \infty$ ,

$$\sup_{(x'_j, x'_k) \in \mathcal{N}_\delta^2} |Z_n(x'_j, x'_k)| = O_p\left((nh_j h_k)^{-1/2} \sqrt{\log n}\right).$$

## Bound (3)

### Combining (2) and (3)

$$\begin{aligned}\sup_{(x_j, x_k) \in [0,1]^2} |Z_n(x_j, x_k)| &\leq \sup_{\substack{|x_j - x'_j| \leq \delta \\ |x_k - x'_k| \leq \delta}} |Z_n(x_j, x_k) - Z_n(x'_j, x'_k)| \\ &\quad + \sup_{(x'_j, x'_k) \in \mathcal{N}_\delta^2} |Z_n(x'_j, x'_k)| \\ &= o_p(1).\end{aligned}$$

### Final step.

$$\sup_{(x_j, x_k) \in [0,1]^2} \hat{p}_{j,k}(x_j, x_k) \leq \sup_{(x_j, x_k) \in [0,1]^2} p_{j,k}(x_j, x_k) + o_p(1).$$

Let  $C' = \sup_{(x_j, x_k) \in [0,1]^2} p_{j,k}(x_j, x_k) + 1$ ; then (1) holds with probability tending to 1. □